

ENGINEERING CHANGE NOTE (ECN)

PROJECT: PROJECT-USERTOS (based on PROJECT-ADDMOTOR)

ORIGINATION: Kidd, Head Technical Lead, UMBC

PURPOSE OF CHANGE: Implement FreeRTOS

DOCUMENT NUMBER: ECN4.1

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LEAD ENGINEER:

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TEAM LEAD:

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EXECUTIVE SUMMARY:

This ECN documents the changes required to replace the cyclic executive round-robin scheduler of PROJECT-ADDMOTOR with a FreeRTOS-based implementation. No customer requirements have changed. One new high-level requirement (HL.19) has been added requiring the software to run on an RTOS. The hardware circuit is unchanged. The four software functions (serial input, LED1, LED2, fan) have been refactored into FreeRTOS tasks with appropriate priorities and a mutex protecting shared serial state. New testing requirements T.13–T.17 have been added to validate FreeRTOS operation and confirm all prior tests still pass.

HIGH-LEVEL DESCRIPTION:

Replace cyclic executive scheduler with FreeRTOS tasks to satisfy HL.19 while preserving all existing behavior.

REASON FOR CHANGE:

ECR 4 (issued by Kidd, 4/1/2026) requires implementation of HL.19: the system and application software must be built upon an RTOS to future-proof the design.

CHANGED DOCUMENTS:

PROJECT REPORT (Demory.cmpe311.spring26.project#4):

- High-Level Requirements: Added HL.19 (4/23/2026, Demory — per Kidd ECR 4, 4/1/2026)
- Educational Requirements: Added E.5 (4/23/2026, Demory — per Kidd, 4/1/2026)
- Testing Requirements: Added T.13, T.14, T.15, T.16, T.17 (4/23/2026, Demory)
- Design Pre-requisites: Added Arduino_FreeRTOS library requirement (4/23/2026, Demory)

- Design Overview: Updated to describe FreeRTOS task architecture (4/23/2026, Demory)
- Testing table: Added rows T13–T17, updated T1–T12 with FreeRTOS verification note (4/23/2026, Demory)

ADDED DOCUMENTS:

project4_freertos.ino — New Arduino sketch implementing FreeRTOS task architecture.

REMOVED DOCUMENTS:

project2code.ino — Superseded by project4_freertos.ino. Retained in repository for reference.

REPLACED DOCUMENTS:

Demory.cmpe311.spring26.project#2 replaced by Demory.cmpe311.spring26.project#4 incorporating HL.19 and T.13–T.17.

ANY ADDITIONAL DESIGN CONSIDERATIONS:

FreeRTOS installation:

- Install 'Arduino_FreeRTOS' by Richard Barry via Arduino IDE Library Manager before compiling.

Task architecture:

- Task_Serial: priority 2, stack 256 words — highest priority to prevent missed serial characters
- Task_LED1: priority 1, stack 128 words — uses vTaskDelay for blink timing
- Task_LED2: priority 1, stack 128 words — uses vTaskDelay for blink timing
- Task_Fan: priority 1, stack 128 words — uses vTaskDelay for button debounce

Mutex usage:

- xSerialMutex protects: interval1, interval2, selectedLED, waitingForInterval
- LED tasks read interval1/interval2 without mutex (single-word volatile read is atomic on AVR)

Memory note:

- Arduino Uno R3 has 2KB SRAM. FreeRTOS itself uses ~200 bytes. Total task stacks: $256+128+128+128 = 640$ words (1280 bytes). Monitor for stack overflow in testing.

Hardware: unchanged from PROJECT-ADDMOTOR — MOSFET on pin 3, button on pin 2, LEDs on pins 7 and 8, 9V battery, flyback diode, smoothing capacitor, gate and pull-down resistors.

NOTE: All deleted text is struck through with date. All additions are identified with date stamps.
This ECN reflects changes made on 4/23/2026 by Nathan Demory per ECR 4 issued by Kidd
4/1/2026.

END OF DOCUMENT